

ST BENEDICT'S COLLEGE



SUBJECT	Mathematics	DATE	31 March 2017
GRADE	11	MARKS	55
EXAMINER	Mrs Sillman	MODERATORS	Mrs Povall & Grade 11 teachers
NAME		DURATION	1 hour
CLASS		TEACHER	

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1 & 2	Exponents and Surds	20	
3	Algebra and Equations	17	
4	Number Patterns	18	
		55	

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

%

QUESTION 1**[14 marks]**

This question is to be completed without the use of a calculator. ALL steps must be shown. Answer only will be awarded ZERO.

Simplify:

a) $\left(\frac{2^{-2}a^5}{3a^{-1}b}\right)^{-2}$

(3)

b) $(3^{-1} - 2^{-1})^2$

(3)

c) $\frac{27^x \cdot 9^{x+1}}{3^{5x+3}}$

(4)

$$d) \frac{16^{\frac{3}{4}} \cdot \sqrt{3} \cdot 4^{\frac{1}{2}}}{27^{-\frac{1}{2}} \cdot 8^{\frac{2}{3}}}$$

(4)

QUESTION 2

[6 marks]

Solve the following equations. Show all working.

a) $3^{2x+1} - 3^{2x} = 18$

(3)

b) $\sqrt[3]{x^2} = 25$

(3)

QUESTION 3**[17 marks]**

- a) Show by completing the square that the function $2x^2 - 12x + 22$ has a minimum value of 4. Explain why you know it is a minimum value.

(5)

- b) Solve for x by completing the square. Leave your answer in simplest surd form.

$$2x^2 + 4x - 8 = 0$$

(5)

- c) Show that there are two points of intersection between the following functions by solving for x and y :

$$y = 2x^2 - 2x - 8 \text{ and } y = 5x - 2$$

(7)

QUESTION 4

[18 marks]

- a) The general term of $5 \quad 12 \quad 29 \quad 48 \quad 77 \dots$ is $T_n = 3n^2 + 2$
Show by using a minimum of TWO calculations whether you agree with the statement or not. Explain your conclusion.

(3)

b) Consider the sequence:

$$\frac{2}{3} \quad \frac{1}{3} \quad \frac{1}{6}$$

- (i) Extend the sequence by 3 terms
- (ii) Determine the general term
- (iii) Determine T_{10}

c) Consider the quadratic pattern:

-9 -6 1 x 27

- (i) Give the value of the second difference
- (ii) Find the value of x
- (iii) Determine which term in the sequence has a value of 397

EXTRA PAPER FOR WORKING/CHECKING